Detecting Deepfake Misinformation on Social Media

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1. Introduction

Deepfakes are a form of synthetic or manipulated media generated using AI that can convincingly depict real people saying or doing things that never occurred. Deepfake generation is accessible to non-experts, this effectively "democratises" misinformation [4]. They are increasingly used in social media misinformation campaigns, seriously threatening public trust in evidence, while the speed of spread and the realities of real-time moderation make harm prevention difficult. We focus our efforts on a single research question:

How can education and detection methods help people to resist and identify social media deepfake misinformation?

A broader contextual question, "How do deepfakes contribute to misinformation across social media platforms, and what are the key social and ethical consequences?", is used to frame the stakes.

We propose that a layered socio-technical defence is necessary for effective resistance. Education and detection cannot individually mitigate the harms of social media deepfake misinformation. Instead, an effective People-Centred defence should combine:

1. **Content provenance** through user-facing decision support such as AI warnings, fact check labels, and poster context (e.g., an account's age and country of origin).
2. **Robust detection** that can withstand social media platform constraints such as compression and filtering.
3. **Education** that builds resilience to misinformation before exposure.

2. Rationale

Why social media deepfake misinformation matters in People-Centred AI.

The social and ethical consequences are widespread and incredibly harmful. For example, deepfakes are increasingly used by state-sponsored actors as a tool in geopolitical misinformation campaigns, such as in the Russian-Ukraine war (on both sides) and recently in Pakistani propaganda [10]. Further, individuals may be targeted through fake non-consensual deepfake content, potentially threatening their reputation.

Social platforms intensify these harms, allowing deepfake misinformation to flourish. Short-form, affective content encourages rapid sharing over content legitimacy, and recommender systems can repeatedly expose users to the same claims. The Illusory Truth Effect states that individuals perceive claims as truer if previously exposed to them, regardless of their cognitive ability. Further, most individuals cannot reliably identify deepfakes, even after raising awareness, yet believe they can [2]. Platform constraints,

such as content filtering and compression, pose additional challenges for deepfake mitigation. For example, lab-trained detectors often degrade when used on deepfakes found on social training setups should consider these constraints [6] Watermarks should withstand common transforms, and may be imperceptible (Google's SynthID [6]), but are not implemented across all Generative AI platforms.

3. Literature Review

3.1. Theme 1: Technical Detection and Limitations

As generative tools have become increasingly accessible through consumer-facing applications and open-source frameworks, the production of high-quality deepfakes has shifted from a niche technical capability to a widely available practice, normalizing the use and spread of misinformation [4].

In response, a substantial body of research has focused on automated deepfake detection. Detection approaches typically analyse spatial artefacts (e.g., inconsistencies in facial texture or lighting), temporal irregularities (e.g., unnatural motion patterns), or multimodal inconsistencies between audio and visual streams. Recent state-of-the-art models, such as universal detectors capable of identifying multiple manipulation types, report high accuracy on benchmark datasets [1]. However, these systems are predominantly developed and evaluated under controlled laboratory conditions, which limits their reliability when deployed in real-world environments such as social media.

State-of-the-art deepfake detection methods now primarily use neural networks trained on large datasets of real and synthetic media. These models learn statistical patterns that differentiate authentic content from manipulated material rather than relying on manually defined rules. Recent detectors aim to generalise across multiple manipulation techniques rather than targeting a single deepfake method. For example, universal detection models such as UNITE are designed to detect a wide range of synthetic video content, including face swaps, reenactments, and fully AI-generated videos, reflecting the diversity of deepfakes circulating online [1]. In controlled experimental settings, such systems often report high accuracy, suggesting strong potential for automated moderation. However, a recurring finding in the literature is that detection performance declines substantially when these systems are deployed in real-world social media environments. Social media platforms routinely apply compression, resizing, filtering, and re-encoding to uploaded content, which alters the statistical properties that detection models rely on. Deepfake detectors trained on clean, high-quality datasets experience significant accuracy degradation when evaluated on videos processed to mimic social

media pipelines [9]. This gap between laboratory conditions and real-world deployment represents a critical limitation of current technical detection approaches.

3.2. Theme 2: Education and Overconfidence in Human Detection Ability

Research consistently demonstrates that humans are remarkably poor at detecting deepfakes, even when aware of the threat. Köbis et al. (2021) [2] conducted a seminal study revealing that people cannot reliably detect deepfakes yet believe they possess this capability. Their research found that participants consistently fail to detect deepfakes at above-chance levels, even after explicit warnings or prior exposure. The participants remain overconfident in their detection abilities, a phenomenon that exacerbates vulnerability rather than mitigating it. This persistent inability highlights a fundamental mismatch between human perceptual capabilities and the sophistication of modern synthesis techniques.

In more recent literature, [5]iProov revealed that only 0.1% of people could correctly distinguish between real and fake content, despite 60% believing they were proficient at detecting deepfakes. This overconfidence creates vulnerability, as individuals who believe they can identify manipulation are less likely to verify information through secondary sources. This study also raises an alarming issue that about 30-40% of older generation have never even heard of the term deepfake. This raises some serious concerns and questions about implementation of methods across all age groups and societies for raising awareness against the effects and detection of deepfake methods. Furthermore, a large group of people also show obvious ignorance for reporting when they come across deepfakes on social media, making moderation an incredibly difficult task for platforms.

Psychological mechanisms amplify the impact of deepfake misinformation. The Illusory Truth Effect (ITE), where repeated exposure to false information increases perceived truthfulness plays a particularly insidious role. A study conducted across eight countries demonstrates that individuals who had previously encountered deepfakes were more likely to perceive the misinformation as accurate than those exposed for the first time [3]. Critically, this pattern held regardless of cognitive ability levels, political orientation, or the political versus non-political nature of the deepfakes. Research shows that knowledge does not protect against the illusory truth effect, individuals who know the correct answer can be swayed to believe otherwise through repetition [7]. Familiar information feels easier to process and hence this cognitive ease makes it easier to misclassify deepfakes the more you are exposed to them.

Educational approaches aim to build critical evaluation skills and resilience against misinformation. While media literacy demonstrates measurable benefits, education alone

cannot solve the deepfake problem. Multiple factors limit the efficacy of awareness campaigns. First, the sophistication of deepfakes increases faster than human detection capabilities improve. A UNESCO report states that we are approaching a "synthetic reality threshold" beyond which humans cannot reliably distinguish authentic from fabricated media without technological assistance, regardless of training [8]. Although education-based results show some improvements, these gaps underscore the need for interdisciplinary research that combines technical robustness with social, ethical, and behavioural insights.

4. Research and Methodology

This study employs a systematic literature review methodology to investigate the role of education and detection methods in combating deepfake misinformation on social media. Literature searches were performed across multiple academic databases, including Google Scholar, IEEE Xplore, ACM Digital Library, PubMed, and arXiv, to capture a broad range of peer-reviewed articles, preprints, and reports published between 2015 and 2025. Inclusion criteria focused on English-language sources that addressed deepfake generation, detection, psychological impacts, or mitigation strategies in social media contexts, excluding purely technical papers without socio-ethical discussions or those unrelated to misinformation. Gray literature, such as reports from UNESCO and industry sources like iProov, was incorporated to provide practical insights, also explored some deepfake examples on X (formerly twitter) to retrieve real-time examples of incidents, such as geopolitical misinformation campaigns referenced in [10].

The review initially focused on literature examining educational and awareness-based interventions, including media literacy, user warnings, and public knowledge of deepfakes. Limitations of educational methods were identified through repeated findings across psychological and behavioural studies showing low human detection accuracy, overconfidence, and susceptibility to cognitive biases such as the illusory truth effect. In addition to the broad systematic search on educational methods, we conducted targeted in-depth reviews of select technical papers to assess state-of-the-art detection methodologies and their real-world applicability. For instance, we examined the UNITE detector detailed in Kundu et al. (2024) [1], which proposes a universal synthetic video detector capable of identifying partial manipulations (e.g., face or background swaps) and fully AI-generated content without relying on face-centric features. This focused review informed our gap analysis by highlighting how such detectors could integrate with educational strategies in hybrid approaches.

The curated review of technical and educational approaches along with relevant articles and live circulating examples of deepfakes on social media from formed the base of this essay.

5. Critical Analysis and Evaluation

The literature review shows that neither technical deepfake detection nor methods like education-based awareness campaigns are sufficient by themselves to effectively mitigate deepfake misinformation on social media. These limitations arise from the interaction between human cognitive vulnerabilities and various technological constraints within the social media platforms. This highlights the need for strategies that can effectively address both content exposure and user susceptibility.

5.1. Limitations of Technical Detection Methods

State of the art models for deepfake detection show strong performance in identifying manipulated media and misinformation with high accuracy. These models include universal and multimodal detectors which utilise the spatial artefacts, temporal inconsistencies and various other cross-modal cues. From a technical perspective, these models provide clear advantage as they are scalable, automated and can reduce user exposure to deepfake content at a platform level.

However, the literature consistently mentions a significant gap between performance on selected datasets and real-world deployment. Social media platforms have constraints like compression, re-encoding, resizing, and sharing, all of which can degrade the quality of the content thereby degrading the quality of the visual and temporal artefacts on which the detection models rely upon. With rapid content dissemination on social media platforms, this gap reflects a broader limitation in current evaluation techniques, which prioritise accuracy on curated benchmark datasets rather than testing the robustness under platform-specific constraints.

Additionally, technical detection methods can be vulnerable to malicious actors who can deliberately change the content to not get detected. Thus, the technical detection methods alone cannot provide a comprehensive or reliable solution to deepfake misinformation on social media because while these methods are effective at reducing the manipulated content, they do not address how users interpret the content that bypasses the automated filters.

5.2. Limitations of Educational and Awareness-based Approaches

Improving users' ability to evaluate the online content can be done through education and awareness campaigns. Due to the increase in deepfake content the authenticity of digital

media is increasingly questioned currently. Media literacy initiatives, and public awareness efforts are designed to promote scepticism towards digital content among users. However, literature suggests that education-based approaches face a lot of significant limitations. Empirical studies like Köbis et al. [2] demonstrate that users are not only poor at detecting deepfakes, but also consistently overestimate their detection abilities. This overconfidence reduces scepticism and discourages the act of verification of the content, which increases the vulnerability rather than mitigating the risk.

Human Psychological mechanisms further complicate this issue. The Illusory Truth Effect shows that repeated exposure to false information increases its credibility, and this phenomenon is further amplified on social media platforms. Many studies have shown that this effect is observed across all different demographics and cognitive groups. UNESCO has argued that advances in synthetic media are reaching to a point where humans cannot reliably tell the difference between authentic and fake content without technological assistance.

5.3. Comparative Evaluation

The limitations of technical detection systems and education-based approaches are complementary rather than independent. Detection models and systems target the availability of the deepfake content by removing and flagging the manipulated content, whereas education-based approaches address user susceptibility by encouraging scepticism and critical thinking. When examined together, the literature demonstrates that each approach addresses a different side of the same problem while leaving critical gaps.

Technical systems can reduce a lot of exposure to the manipulated content but fail to address how users interpret and trust the content, which evades detection. Conversely, education campaigns tend to assume that the users are attentive and rational despite the evidence of cognitive bias and overconfidence. Social media platforms further fuel these limitations by promoting short form content, rapid consumption, repetition, targeted algorithms. As a result, neither approach is sufficient on its own to mitigate the core issue of societal and ethical harm caused by deepfake misinformation.

This evaluation suggests that deepfake misinformation on social media platforms must be considered as a socio-technical problem, which arises from the interaction between algorithms, social media platforms, and human cognition, rather than as an issue that can be resolved through just technical or educational solutions.

5.4. People Centred Implications

From a people-centred AI perspective, the findings indicate that effective deepfake detection on social media must consider the human behaviour and ethical considerations along with technically robust detection. A combined strategy might consist of a robust technical model along with user-facing interventions such as transparent labelling, contextual warnings and targeted education aimed at supporting informed decision-making among users.

Importantly, the success of this approach will not be solely measured in terms of detection accuracy, but also in terms of reduced harm, improved user trust, and informed decision making among users. By addressing both technological limitations and human vulnerabilities, people-centred strategies can provide a more realistic and ethically grounded foundation for responsible AI deployment on social media platforms.

6. Conclusion

This essay examines the performance of deepfake detection methods and education-based awareness campaigns in mitigating deepfake misinformation on social media. The findings show that real-world conditions, such as platform-specific processing and rapid sharing of content, affect the technical accuracy of new detection technologies. At the same time, humans remain vulnerable to cognitive bias and often overestimate their ability to identify manipulated content. As a result, education and awareness initiatives alone are not sufficient to counter deepfake misinformation, highlighting the need for mitigation strategies that account for both technological limitations and realistic user behaviours.

Overall, the literature points to the need for a combined, people-centred approach that brings technical detection tools and education together. Instead of expecting users to identify deepfakes on their own, social media must take responsibility by using detection tools to limit the spread of deepfake content, add warning signs, and make reporting easier. Education should be focused more on encouraging users to verify the source of the content and think carefully before sharing, rather than focusing less on teaching people to spot deepfakes.

Future research should prioritise improving the robustness of detection models by evaluating them under realistic platform conditions. There is also a need for interdisciplinary research that combines technical, psychological, and social perspectives to better understand how user behaviour and platform design help in amplifying

misinformation. Addressing these gaps is important for developing effective strategies for maintaining trust in digital media as deepfake technologies continue to evolve.

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